

Exact Sequences, Projective, Injective, and Flat Modules

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1 Exact Sequences

Given R -modules A, C , find all modules B such that the sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is exact.

Claim: $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$ is exact iff $0 \rightarrow \alpha(X) \rightarrow Y \rightarrow Y/\ker \beta \rightarrow 0$ $(\star)$ is exact.

Proof: Suppose $X \rightarrow Y \rightarrow Z$ is exact. $(\star)$ is exact at $\alpha(X), Y/\ker \beta$. Note that $\beta\alpha = 0 \rightarrow \alpha(X) \leq \ker \beta$. Then the maps compose to 0 at Y . Finally, suppose $y \in \ker \beta$. Then $y = \alpha x$ for some $x \in X$, so $(\star)$ is exact at Y .

Now suppose $0 \rightarrow \alpha(X) \rightarrow Y \rightarrow Y/\ker \beta \rightarrow 0$ is exact. Let $x \in X$. Since $(\star)$ composes at 0 at Y , we have $\alpha(X) \leq \ker \beta$, so $\beta\alpha X = 0$. Suppose $\beta y = 0$. Then $y \in \ker \beta$, so by exactness of $(\star)$ at Y , there exists $x \in X$ such that $\alpha x = y$, so $y \in \operatorname{im} \alpha$, ie $X \rightarrow Y \rightarrow Z$ is exact.

We can always form the short exact sequence $0 \rightarrow A \rightarrow A \oplus C \rightarrow C \rightarrow 0$. An extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$ is $0 \rightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$.

If $\varphi : B \rightarrow C$ is a homomorphism, we have the exact sequence $0 \rightarrow \ker \varphi \rightarrow B \rightarrow \operatorname{im} \varphi \rightarrow 0$. Take M to be generated by S , $F(S)$ to be the free module on S . Then we have $0 \rightarrow K \rightarrow F(S) \rightarrow M \rightarrow 0$.

$0 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_2 \oplus K_4 \rightarrow K_4 \rightarrow 0$. $0 \rightarrow \mathbb{Z}_2 \xrightarrow{\varphi} \mathbb{Z}_2 \oplus \mathbb{Z}_4 \xrightarrow{\psi} K_4 \rightarrow 0$, where $\varphi x = (0, x)$ and $\psi(x, y) = (x, 2y)$. Indeed, $1 \mapsto (0, 2) \mapsto (0, 0)$. If $\psi(x, y) = 0$, then $x = 0$ and $y = 0, 2$ and $(0, 2) = \varphi(1)$. We have $\mathbb{Z}_2 \oplus K_4 \cong \mathbb{Z}_2^{\oplus 3}$, which is not isomorphic to $\mathbb{Z}_2 \oplus \mathbb{Z}_4$.

Other extensions of K_4 by $\mathbb{Z}_2$ are

$$\begin{aligned} 1 \rightarrow \mathbb{Z}_2 \rightarrow D_4 \rightarrow K_4 \rightarrow 1 \\ 1 \rightarrow \mathbb{Z}_2 \rightarrow Q_8 \rightarrow K_4 \rightarrow 1 \end{aligned}$$

where $\mathbb{Z}_2$ is identified with the respective centers of the groups and K_4 is the quotient by the center.

Claim: Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0, 0 \rightarrow A' \rightarrow B' \rightarrow C' \rightarrow 0$ be short exact sequences with $\alpha : A \rightarrow A', \beta : B \rightarrow B', \gamma : C \rightarrow C'$ such that (α, β, γ) is an isomorphism. Then $(\alpha^{-1}, \beta^{-1}, \gamma^{-1})$ is an isomorphism in the reverse direction.

Proof: We have $\beta\varphi = \varphi'\alpha$ and $\gamma\psi = \psi'\beta$. Then $\varphi\alpha^{-1} = \beta^{-1}\varphi'$ and $\psi\beta^{-1} = \gamma^{-1}\psi'$, so $(\alpha^{-1}, \beta^{-1}, \gamma^{-1})$ is an isomorphism.

Claim: Let B, B' be equivalent extensions of C by A via $\beta : B \rightarrow B'$. Then β^{-1} is an equivalence.

Proof: We have $\beta\varphi = \varphi'$ and $\psi'\beta = \psi$. Then $\beta^{-1}\varphi' = \varphi$ and $\psi\beta^{-1} = \psi'$.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{n} & \mathbb{Z} & \xrightarrow{\pi} & \mathbb{Z}_n \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\iota} & \mathbb{Z} & \xrightarrow{\pi'} & \mathbb{Z}_n \longrightarrow 0 \end{array}$$

2 Noetherian Rings

Let $\text{Int}(\mathbb{Z}) = \{f \in \mathbb{Q}[x] : f(\mathbb{Z}) \subseteq \mathbb{Z}\}$. Note that $\mathbb{Z}[x] \subset \text{Int}(\mathbb{Z})$ since $x(x-1)/2 \in \text{Int}(\mathbb{Z}) - \mathbb{Z}[x]$. In general, let $f_n(x) = \binom{x}{n} \in \text{Int}(\mathbb{Z})$. I claim that $\{f_n : n \in \mathbb{Z}_{\geq 0}\}$ is a $\mathbb{Z}$ -basis of $\text{Int}(\mathbb{Z})$. Note that $f_0 = 1$.

The polynomials f_i have distinct degrees, so are independent over $\mathbb{Z}$. Suppose $f \in \text{Int}(\mathbb{Z})$ is nonzero. I claim that $f \in \langle f_n : n \in \mathbb{Z}_{\geq 0} \rangle$ by induction. Let f have degree n and leading coefficient a . Then $n!a = \sum_{k=0}^n \binom{n}{k} (-1)^k f(n-k) \in \mathbb{Z}$. Consider $g = f - (n!a)f_n$, which has degree at most $n-1$. Since $f, f_n \in \text{Int}(\mathbb{Z})$ and $n!a \in \mathbb{Z}$, we have that $g \in \text{Int}(\mathbb{Z})$. By the IH, we're done.

Claim: The following are equivalent:

- (i). Let R be a commutative ring. R is said to be Noetherian if every ideal $I \leq R$ is finitely generated.
- (ii). Let $\{I_n : n \in \mathbb{Z}^+\}$ be an ascending sequence of ideals, ie $I_n \leq I_{n+1}$. Then there exists $N \in \mathbb{Z}^+$ such that $I_n = I_N$ for all $n \geq N$.

(iii). Every nonempty collection $S \neq \emptyset$ of ideals contains a maximal element with respect to inclusion.

Proof: (i) implies (ii). Let $\{I_n\}$ be an ascending sequence of ideals. Let $I = \bigcup_{n \in \mathbb{Z}^+} I_n$, which I claim is an ideal. Indeed, $0 \in I$. If $x, y \in I$, then $x \in I_n, y \in I_m$ for some $n, m \in \mathbb{Z}^+$. Let $k = \max(n, m)$, so $x, y \in I_k$ and $x - y \in I_k \leq I$. Let $r \in R$ be arbitrary. Then $rx \in I_n \leq I$, so indeed $I \leq R$.

Let $I = (a_i : i \in [n])$ for some finite number of $a_i \in R$. For all $i \in [n]$, we have $i \in I$, so $i \in I_{n_i}$ for some $n_i \in \mathbb{Z}^+$. Let $N = \max_{i \in [n]} n_i$, which exists since $[n]$ is finite. Then $a_i \in I_N$ for all $i \in [n]$. It follows $I \leq I_N \rightarrow I_N = I$. For $n \geq N$, we have $I_N \leq I_n \leq I = I_N \rightarrow I_n = I_N$, as desired.

(ii) implies (iii). Since $S \neq \emptyset$, pick $I_1 \in S$. Assume I_n has been picked. If I_n is maximal in S , we're done. Else, pick $I_{n+1} \in S$ such that $I_n < I_{n+1}$. By induction, we construct a strictly ascending chain of ideals $\{I_n : n \in \mathbb{Z}^+\}$, absurd.

(iii) implies (i). Let $I \leq R$ be an ideal. Let $S = \{J \leq I : J \text{ is finitely generated}\}$. We can take $J = 0$ to be generated by $\emptyset$, so $S \neq \emptyset$. Let $M \in S$ be maximal. Suppose FTSOC $M < R$. Pick $a \in R - M$. Then $(a, M) > M$ is finitely generated, contradicting maximality.

Consider the original problem. Let $I_n = (f_i : i \in [n])$. I claim that $f_p \notin I_{p-1}$ for all primes p . Suppose

$$\binom{x}{p} = \sum_{k=1}^{p-1} g_k \binom{x}{k}$$

for some $g_k \in \text{Int}(\mathbb{Z})$. Then $\binom{x}{k}$ is a $\mathbb{Z}$ -linear combination of terms of the form $\binom{x}{a} \binom{x}{b}$, where $a \in [p-1]$ and $b \in \mathbb{Z}_{\geq 0}$. Note that

$$\binom{x}{a} \binom{x}{b} = \sum_{l=0}^{\min(a,b)} \frac{1}{a!b!} \binom{a}{l} \binom{b}{l} l!(x)_{a+b-l} = \sum_{l=0}^{\min(a,b)} \binom{a+b-l}{a-l, b-l, l} \binom{x}{a+b-l}.$$

We want $a + b - l = p$ (since the f_i is a $\mathbb{Z}$ -basis of $\text{Int}(\mathbb{Z})$). If $l \neq 0$, then $p \mid \binom{a+b-l}{a-l, b-l, l}$. Else, $l = 0$, so $a + b = p$, so $p \mid \binom{a+b}{a, b}$ since $a \in [p-1]$. Hence, $1 \equiv 0 \pmod{p}$, absurd.

Let p_i denote the i th prime. We have the strictly ascending sequence of ideals $(I_{p_n} : n \in \mathbb{Z}^+)$ (this is an infinite sequence since there are infinitely many primes), so $\text{Int}(\mathbb{Z})$ is not Noetherian.

Claim: Let R be Noetherian. Every surjective ring homomorphism $f : R \rightarrow R$

is injective, ie an isomorphism.

Proof: Suppose $J \trianglelefteq S$ and $f : R \rightarrow S$ is a homomorphism. Then $f^{-1}(J) \trianglelefteq R$. Let $r \in R$ and $x \in f^{-1}(J)$. Then $f(rx) = f(r)f(x) \in J$, so $rx \in f^{-1}(J)$. Note $0 \in f^{-1}(J)$. If $x, y \in f^{-1}(J)$, then $f(x-y) = f(x) - f(y) \in J$, so $x-y \in f^{-1}(J)$. Hence, $f^{-1}(J) \trianglelefteq R$.

Let f have its original meaning. Let $I_1 = \ker f \trianglelefteq R$. Define $I_{n+1} = f^{-1}(I_n)$ for $n \geq 1$. I claim that $I_n \trianglelefteq I_{n+1}$. Indeed, define $I_0 = 0$. I claim by induction $I_n \trianglelefteq I_{n+1}$ for all $n \geq 0$. We have $0 \leq \ker f = I_1$. Assume the result for n . Then $f^{-1}(I_n) \trianglelefteq f^{-1}(I_{n+1}) \rightarrow I_{n+1} \trianglelefteq I_{n+2}$. $I_n \trianglelefteq R$ by induction as well.

Let $\{I_n\}_{n \in \mathbb{Z}^+}$ stabilize at $N \in \mathbb{Z}^+$. Then $f^{-1}(I_N) = I_N$. Since f is surjective, we have $ff^{-1} = 1$. If $N > 0$, write $f^{-1}f^{-1}(I_{N-1}) = f^{-1}(I_{N-1})$. Applying f to both sides gives $f^{-1}(I_{N-1}) = I_{N-1}$. By induction, $f^{-1}(I_0) = I_0 = 0$, ie f is injective.

Claim: Let R be a Noetherian integral domain. Then every nonzero nonunit $r \in R$ can be factored into a product of irreducibles.

Proof: $s \in R$ is said to be an irreducible if $s \neq 0, s \notin R^\times$ and there do not exist nonunit $t_1, t_2 \in R$ such that $s = t_1 t_2$. Let $r \in R$ be nonzero nonunit.

I claim that there exists an irreducible factor of r . Let $r_0 = r$. Suppose $r_n \mid r$ is defined. If r_n is irreducible, we're done. Else, pick $r_{n+1} \mid r_n$ such that r_{n+1} is nonzero nonunit and r_{n+1}, r_n are not off by a unit. Then $r_{n+1} \mid r$ is nonzero nonunit. Let $I_n = (r_n)$. Since $r_n \mid r_{n+1}$, we have $I_n \leq I_{n+1}$. If $I_n = I_{n+1}$, then $r_n \mid r_{n+1}$, which implies r_n, r_{n+1} are off by a unit, absurd. Hence, the sequence of ideals (I_n) is strictly increasing, absurd.

Let $s_0 = r$. Suppose s_n is defined and is nonzero nonunit. Let a_n be an irreducible factor of s_n and let $s_n = a_n s_{n+1}$. If s_{n+1} is a unit u , then $r = u \prod_{i \in [n]} a_i$. Else, let $I_n = (s_n)$. Since $s_{n+1} \mid s_n$, we have the ascending chain of ideals $I_n \leq I_{n+1}$. Since R is Noetherian, there exists $N \in \mathbb{Z}^+$ such that $I_N = I_{N+1}$, so s_N, s_{N+1} are off by a unit. However, s_N, s_{N+1} are off by a nonzero nonunit a_N , absurd.

Claim: If R is Noetherian, then so is R/I for every $I \trianglelefteq R$.

Proof: Suppose $J \trianglelefteq R/I$ is an ideal. Let $\pi : R \rightarrow R/I$ be the projection map. We have $\pi^{-1}(J) \trianglelefteq R$ is finitely generated. Then $\pi\pi^{-1}(J) = J$ is finitely generated.